

Title

Glucose-lowering effects of acute cold-induced thermogenesis in individuals with type 2 diabetes but not prediabetes: a randomized crossover trial

Abstract

Cold-induced metabolic activation could theoretically enhance glucose clearance, though effects in glucose-dysregulated populations remain unexplored. In a randomized crossover design, 24 participants (12 with prediabetes, 12 with established type 2 diabetes), aged 40-75, underwent two controlled cold-exposure sessions using individualized temperature protocols. Sessions were randomly ordered and participants were blinded to exposure intensity. Continuous glucose monitoring tracked interstitial glucose levels over 24-hour periods before and after each exposure. Shivering intensity and resting metabolic rate increases were confirmed via physiological measurement. In prediabetic participants, neither cold-exposure protocol altered 24-hour glucose patterns. Conversely, in type 2 diabetes participants, mild cold exposure (but not the more intense protocol) produced a small yet statistically significant reduction in mean 24-hour glucose levels and fasting glucose values, accompanied by increased time spent within optimal glucose ranges. These metabolic improvements did not correlate with acute energy expenditure during cold exposure, suggesting effects were dependent on baseline metabolic dysfunction rather than sheer caloric mobilization. Findings suggest cold-induced thermogenesis may modulate glucose handling specifically in advanced diabetes, with potential implications for adjunctive non-pharmacological diabetes management strategies.